THE EFFECT OF PORT GEOMETRY ON THE FREQUENCY RESPONSE OF DESKTOP SPEAKERS

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ABSTRACT

The low-frequency response of ported speaker enclosures is modeled by the Helmholtz resonator and lumped-parameter models, which predict that port geometry controls the resonant frequency and quality factor of the system. This study experimentally evaluated the effects of port length and port diameter on the average output level and response uniformity (RMS deviation) of a 3D-printed desktop speaker enclosure over a bass frequency range 126 Hz to 500 Hz. Twenty-one port configurations spanning lengths 0 mm to 180 mm and diameters of 0 mm to 70 mm were tested using a logarithmic sweep sine and far-field microphone measurements. Increasing port diameter was found to significantly increase the average volume and decrease the RMS deviation of the response, consistent with the theoretical prediction that lower quality factors result in flatter resonance peaks that more evenly supplement driver response. Increasing port length was found to significantly decrease the average volume, though physically insignificantly in magnitude, and did not significantly affect RMS deviation, consistent with the theoretical prediction that higher quality factors result in sharper resonance peaks for which the location of the resonant frequency relative to the bass roll-off drives the effect on response. The experiment validated and generalized high-level trends between port geometry and response volume and uniformity through the quality factor.

Keywords: Helmholtz resonator, spectral flatness, bass-reflex, porting, speakers

INTRODUCTION

Speaker sound quality is generally quantified by two parameters in the context of music, those being the uniformity of frequency response and the magnitude of sound pressure levels, equivalent to volume [1]. Amplifiers send a signal to speaker drivers, or cones, which oscillate according to the voltage and frequency from the amplifier through electromagnetic forces. In order to produce low-frequency sound, the cone must oscillate in large excursions, reducing efficiency and lowering acoustic output. In order to reduce the load on the driver, a port, which is a tube that connects the internal air volume of the enclosure to the environment, can be added to the enclosure. The addition of a port creates a resonant air mass that stores energy, radiates sound, and reduces load on the driver, thereby improving the performance of the speaker at lower frequencies. In order to optimize the low-frequency response of a ported enclosure, the diameter and length of the port must be changed so that the air mass in the port resonates at a particular frequency, which is chosen according to the driver and enclosure parameters. Improperly designing a port can result in a response with a resonant frequency which does not effectively supplement the normal bass response of the enclosure, or overamplifies the response at the particular frequency, producing unwanted noise.

Ported enclosures, shown in Figure 1, also referred to as bass-reflex systems, are typically analyzed through the lumped-parameter model wherein the enclosure volume acts as an acoustic compliance (behaving similarly to a spring) and the port as an acoustic mass (behaving similarly to a physical mass). The interaction between the acoustic mass and acoustic compliance governs low-frequency response. However, this model does not account for parameters such as specific enclosure geometry, specific port geometry, enclosure material, among other factors. Thus, experimental evaluation of the system response to port geometry is necessary in assessing the reliability of the model and common approximations used today in speaker design. The results of the study are compared against model predictions to assess accuracy and identify potential sources of discrepancy.

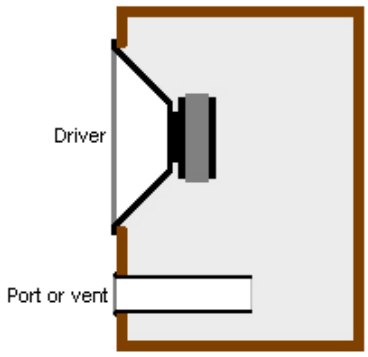


Figure 1: Diagram of a ported speaker enclosure [2].

THEORY

Ported speaker enclosure behavior is commonly analyzed through lumped-parameter models and Helmholtz resonator theory which outline parameters such as the resonant frequency of the port, physical equivalents in the lumped-parameter model, and the total quality factor. These models are limited in that they do not account for particular geometric features, un-uniform enclosure pressure, and other factors present in a speaker.

THEORY OF PORTING ENCLOSURES

At lower frequencies, driver cones must oscillate with larger amplitudes in order to produce the same volume, as larger wavelength sound causes a reduction in sound radiation efficiency [3]. Although larger cone excursions initially compensate for the reduction in sound radiation efficiency, electro-mechanical factors, such as the motor force and maximum excursion range ultimately constrains the output, causing lower volumes at lower frequencies, called roll-off. In sealed enclosures, the sealed volume of air adds stiffness to the system as the cone oscillates, which increases the restoring force on the cone, amplifying the effects of roll-off. The introduction of a port creates a resonant mass which redistributes energy from the enclosure into the port, particularly at lower frequencies. In practice, the port is tuned relative to the driver's mechanical characteristics (Thiele-Small parameters) such that the resonant frequency of the port complements the driver's natural roll-off. In this study, however, the driver is held fixed and the port geometry is varied, isolating the port's acoustic behavior.

The port acts as an acoustic mass which oscillates according to the geometry of the port, and the changes in pressure in the enclosure. The air in the enclosure acts as a spring, which is affected both by motion of the cone and the oscillating air in the port. Similar to a mass-spring system, the acoustic mass-compliance interaction resonates with frequency [4].

$$f_p = \frac{1}{2\pi}\sqrt{\frac{S^2}{M_p C_p}} = \frac{c}{2\pi}\sqrt{\frac{S}{VL}} \tag{1}$$

S is defined as the cross section of the port, V is defined as the internal volume of the enclosure, and L is defined as the length of the port. The acoustic mass behaves as a physical mass [5]:

$$M_p = \rho\frac{L}{S} \tag{2}$$

The acoustic compliance behaves as a physical spring with stiffness [5]:

$$C_p = \frac{V}{\rho c^2} \tag{3}$$

The quality factor of the system describes the damping of the system, based on the resonator characteristics [6]. Lower values of Q represent systems which are heavily damped, which at the cost of output, flattens the response.

$$Q \propto \frac{1}{R_{loss}}\sqrt{\frac{M}{C}} \tag{4}$$

These elements defined the lumped-parameter model and the Helmholtz resonator submodel. As the frequency of the cone oscillations approaches the resonant frequency of the port, sound is radiated from the port at a larger volume, thereby supplementing the driver response in the roll-off regime. The coupled acoustic mass and compliance system thus exhibits second-order resonance behavior derived from linearized fluid dynamics. When the port frequency is tuned appropriately, the sound radiated from the port compensates for the low-frequency roll-off without creating an unwanted resonance peak.

Changing the port length directly affects the acoustic mass. The quality factor is directly related to the port length, therefore increasing the port length increases Q, reducing damping and worsening response uniformity.

Changing the port diameter inversely affects the acoustic mass. The quality factor is inversely related to the port diameter, therefore increasing the port diameter decreases Q, and improves response uniformity.

As the resonant frequency falls below the tested bass response frequency range, increasing the resonant frequency generally results in an increase in the average volume over the range. Since resonant frequency increases with port diameter and decreases with port length, the average bass response volume is expected to increase with port diameter and decrease with port length. For port length, this prediction is reinforced by the Q effect, as higher Q concentrates output near the resonance frequency, which falls outside the tested range. For port diameter, the resonant frequency and Q effects reinforce each other. Wider ports raise the resonant frequency toward the tested range while simultaneously broadening the peak, both of which increase average output in the measurement window. The effect of diameter on flatness is less certain, as the rising resonant frequency and broadening peak interact differently depending on how close the resonant frequency is to the lower bound of the tested range.

LIMITATIONS OF CURRENT THEORY

The lumped-parameter model represents the air in the enclosure as a uniform acoustic compliance and the port air column as an acoustic mass, which forms a second-order mass-spring system, which affects the low-frequency response of a bass-reflex system. This approach assumes enclosure pressure is spatially uniform, that acoustic behavior is linear, and with idealized cylindrical port geometry. In practice, however, deviations from these assumptions can affect the system behavior, thus changing the response of the speaker. Many exceptions to these assumptions arise in discrepancies in boundary conditions, such as flanged ports and enclosure edges. These geometric factors, which are difficult to model, create turbulence, or chuffing, as a result of non-uniform pressure distributions. As a result, the theoretical tuning frequency and predicted response drive from the lumped model may differ from measured performance. The results of the study are compared against model predictions to assess accuracy and identify potential sources of discrepancy.

EXPERIMENTAL DESIGN

A 3D-printed desktop speaker enclosure was used as the test platform. The enclosure housed a tweeter and a midrange driver, with an external passive crossover circuit splitting the signal at approximately 2000 Hz. Since all analysis was conducted below 500 Hz, only the midrange driver was relevant to the measurements. An external amplifier drove the system. A sweep sine spanning 20 Hz to 20 kHz, scaled logarithmically, was stored on a USB drive and played through the system for each trial.

ENCLOSURE AND PORT CONFIGURATION

The speaker enclosure lid contained a circular lipped aperture into which interchangeable 3D-printed ports were friction-fit, as shown in Figure 2.

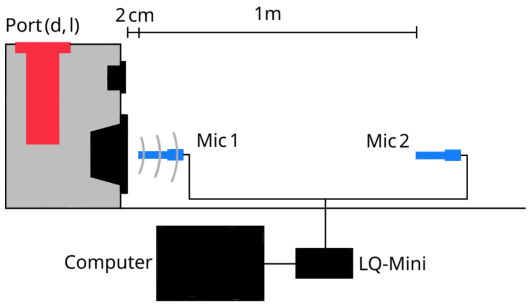


Figure 2: Diagram of the experimental setup. Ported speaker, with a port of diameter d and length l. Near-field mic 2 cm from the cone, far-field mic 1 m from the cone. Data acquisition through an LQ-Mini.

The mating surface between the ports and the lip was lined with PTFE (Teflon) tape to ensure an airtight seal. Ports were fabricated with five diameters – 0 mm, 17.5 mm, 35.0 mm, 52.5 mm, and 70.0 mm – and with lengths – 0 mm, 45.0 mm, 90.0 mm, 135.0 mm, and 180.0 mm. The 0 mm diameter condition corresponds to a sealed enclosure with no port, so that only zero length was physically meaningful, thus there are a total of 21 port geometries, as shown in Table 1.

Table 1: A total of twenty-one port geometries used in the study. A diameter of 0 mm is equivalent to a sealed enclosure, wherein there is no physically meaningful port length.

		Port Diameters [mm]				
		0.0	17.5	35.0	52.5	70.0
Port Len. [mm]	0.0	✓	✓	✓	✓	✓
	45.0		✓	✓	✓	✓
	90.0		✓	✓	✓	✓
	135.0		✓	✓	✓	✓
	180.0		✓	✓	✓	✓

DATA ACQUISITION

A 12-second 20 Hz to 20 kHz logarithmically scaled sweep sine was stored on a USB drive and connected to a 40 W amplifier. The sweep sine allowed for total signal analysis, and the logarithmic scale allowed for high resolution at lower frequencies. The amplifier sent raw audio signal to a crossover with a cutoff frequency of approximately 2000 Hz, which sent high frequencies to the tweeter and bass response frequencies to the midrange.

Two Vernier MCA microphones [Vernier MCA-BTA] were placed at 2 cm and 1 m from the center of the midrange cone in order to measure near-field and far-field responses, respectively. The microphones are rated for 100 Hz to 15 kHz collection. Both microphones were connected to a LabQuest Mini for 12-bit resolution data collection at 10 kHz. Only the far-field mic at 1 m was used for data analysis in order to analyze trends in normal listening conditions, rather than near-field driver effects which are sensitive to local cone effects.

For each of the 21 possible port geometries, data were recorded five times, wherein the sweep sine was played twice per trial and averaged to reduce noise. Between each set of five trials, the port was removed and replaced with another port.

FREQUENCY RESPONSE METRICS

Raw SPL data was exported to MATLAB, where a Fast Fourier Transform (FFT) was applied to the time domain of the recordings and the input sweep sine. Transfer functions were created from the FFTs of the recordings and sweep sine, from which Bode plots were generated over the frequency range 126 Hz to 500 Hz. The five trials were averaged for each configuration.

$$\bar{L} = \frac{1}{500\ \text{Hz} - 126\ \text{Hz}} \int_{126\ \text{Hz}}^{500\ \text{Hz}} \text{SPL}(f)\mathrm{d}f \tag{5}$$

Equation 5 was used to determine the average volume of the response over the defined frequency range.

$$\text{RMS Deviation} = \sqrt{\int_{126\ \text{Hz}}^{500\ \text{Hz}} (f - \bar{L})^2 \mathrm{d}f} \tag{6}$$

Equation 6 was used to determine the RMS deviation of the response over the defined frequency range. A response with higher uniformity is characteristic of a lower RMS deviation, as the RMS deviation is a measure for the degree to which the response differs from a perfectly uniform response at the average volume.

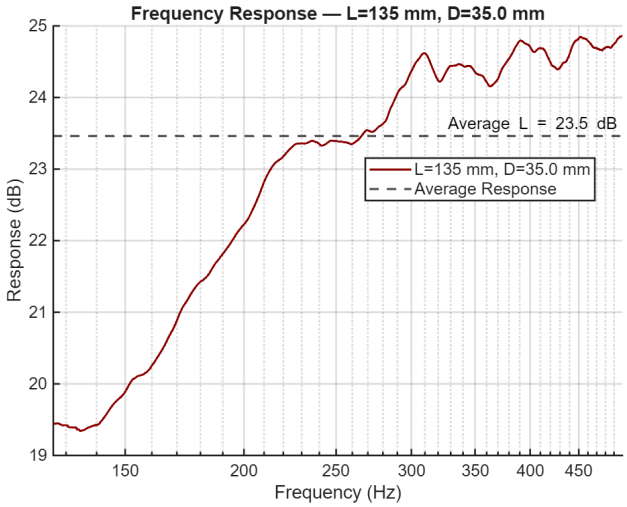


Figure 3: Representative trial with a port of length 135.0 mm and diameter 35.0 mm. The dashed line describes the average volume of the particular response, defined by Eq. 5.

The trial data have been averaged for each port configuration prior to further analysis.

RESULTS AND DISCUSSION

The average output level and RMS deviation were computed for each port configuration across five trials and averaged, and compared against the directional predictions of the lumped-parameter and Helmholtz resonator models developed in Section 2.

EFFECT OF PORT LENGTH

The linear fit of volume relative to the sealed enclosure versus port length yielded a slope of -0.0053 dB/mm, which was statistically significant at the 95% confidence level (95% CI: [-0.0091, -0.0014]), as shown in Figure 4. This trend supports the theoretical prediction that average response volume decreases with port length, as the resonant frequency shifts further below the tested range and higher quality factor concentrates output away from the range. The fit predicted that, for every tested length, volume exceeded the sealed baseline of 23.25 ± 0.43 dB.

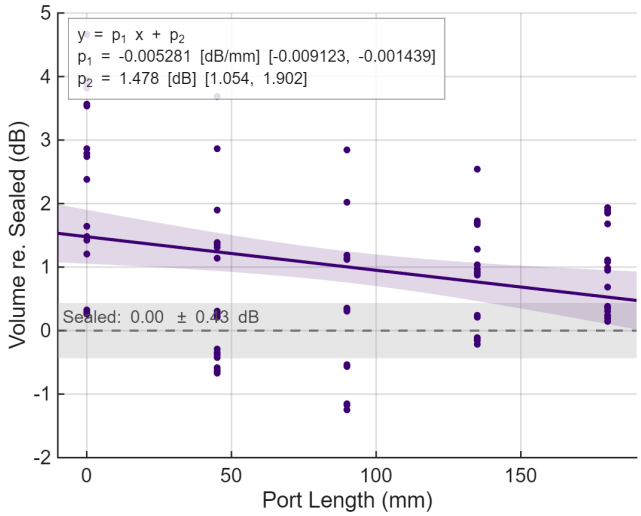


Figure 4: Relative output level as a function of port length, pooled from all port diameters, with a linear fit and a 95% prediction interval.

The linear fit of RMS deviation versus port length yielded a slope of 0.0012 dB/mm, which was not statistically significant at the 95% confidence level (95% CI: [-0.0002, 0.0023]), as shown in Figure 5. The statistically insignificant result neither confirms nor contradicts the theoretical prediction that RMS deviation increases and response uniformity worsens with port length. Notably, the sealed enclosure RMS deviation of 1.86 ± 0.03 dB suggests that porting generally improves flatness relative to a sealed design across the range of tested length.

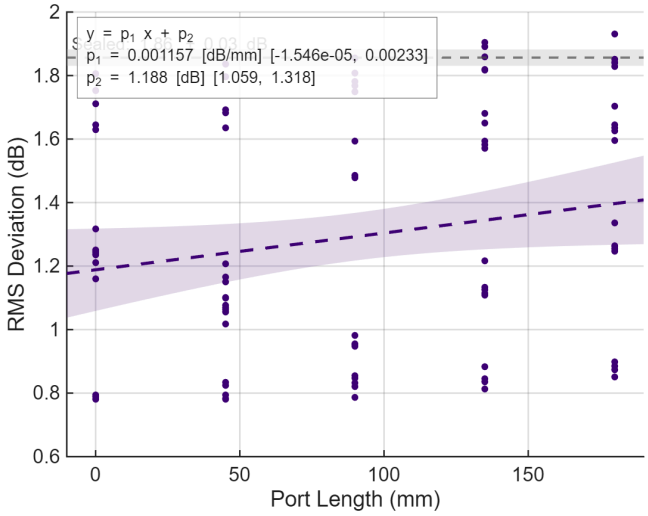


Figure 5: RMS deviation as a function of port length, pooled from all port diameters, with a linear fit and a 95% prediction interval.

EFFECT OF PORT DIAMETER

The linear fit of volume relative to the sealed enclosure versus port diameter yielded a slope of 0.0424 dB/mm, which was statistically significant at the 95% confidence level (95% CI: [0.0326, 0.0521]), as shown in Figure 6. This trend is consistent with the theoretical prediction that average output level increases with port diameter, as wider ports raise the resonant frequency toward the tested range while simultaneously broadening the peak, both of which supplement the response in the tested range. Ported configurations with larger diameters generally exceeded the sealed baseline of 23.25 ± 0.43 dB.

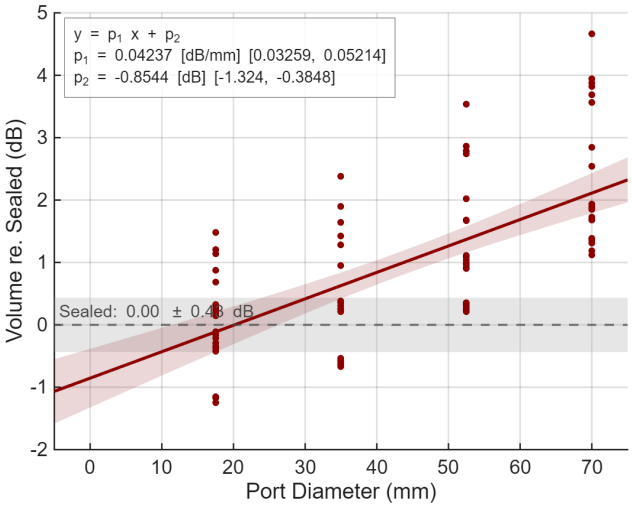


Figure 6: Relative output level as a function of port diameter, pooled from all port lengths, with a linear fit and a 95% prediction interval.

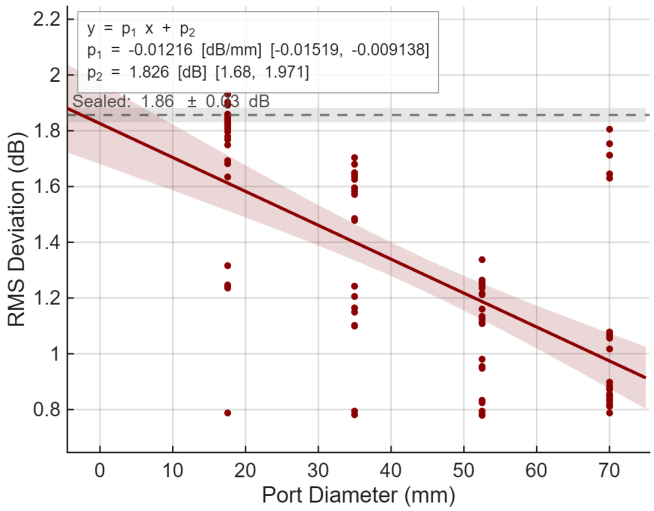


Figure 7: RMS deviation as a function of port diameter, pooled from all port lengths, with a linear fit and a 95% prediction interval.

The linear fit of RMS deviation versus port diameter yielded a slope of −0.0121 dB/mm, which was statistically significant at the 95% confidence level (95% CI: [−0.0152, −0.0091]), as shown in Figure 7. This trend is consistent with the theoretical prediction that RMS deviation decreases with port diameter, as wider ports raise the resonant frequency toward the tested range while simultaneously broadening the peak, which reduces deviation from the average response. Ported configurations with larger diameters generally produced lower RMS deviation than the sealed reference of 1.86 ± 0.03 dB.

LIMITATIONS ON CURRENT THEORY AND EXPERIMENTATION

The Helmholtz resonator model assumes lossless acoustics and idealized cylindrical port geometry. In practice, port losses, boundary layer effects, and radiation resistance all reduce the effective quality factor below the idealized prediction. Thus, the experimental response is flatter than what the model predicts, therefore volume and uniformity metrics are less sensitive to geometric changes than expected. Additionally, the port behavior is governed by an effective length, rather than the true length, and the effective length is dependent on factors such as port-boundary clearance, port end geometry, and other factors which are difficult to quantify without taking additional data. External factors such as port material impermeability and variability in amplifier behavior also compromise the accuracy of experimental data. Vernier MCA microphones are rated for 100 Hz, which is near the 126 Hz lower analysis bound, thus technological limitations affect accurate low-frequency data acquisition. Predicted resonant frequencies ranged from 31.1 Hz to 640.8 Hz, which introduces complications as the behavior of a port largely depends on the location of the resonant frequency relative to the tested range.

Future work could involve enclosures designed so that port geometries have resonant frequencies consistently below the tested frequency range in order to assess the quantitative accuracy of Helmholtz resonator theory and the lumped-parameter model. Testing

enclosures and ports fabricated from different, less permeable materials, and designed to minimize edge effects could further the experiment, as the data would become more representative of the interaction between the port and enclosure, rather than of external factors.

CONCLUSIONS

This study experimentally evaluated the effects of port length and diameter on the average output level and response uniformity of a ported desktop speaker enclosure over a bass frequency range 126 Hz to 500 Hz, assessing whether the Helmholtz resonator and lumped-parameter models provide reliable guidelines for port design.

Three of the four predicted directional trends were statistically significant at the 95% confidence level. Increasing port length significantly decreased average output level at a rate of 0.0053 ± 0.0039 dB/mm, while its effect on RMS deviation was not statistically significant. Increasing port diameter significantly increased average output level at a rate of 0.0424 ± 0.0098 dB/mm and improved response uniformity, with RMS deviation decreasing at a rate of 0.0121 ± 0.0031 dB/mm.

The results confirm that the Helmholtz resonator and lumped-parameter models accurately predict directional trends in bass response, and indicate that port diameter is a more predictable and impactful design parameter than port length over the tested frequency range.

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